

PRECLINICAL QUALIFYING EXAMINATION

AAU, 2k11 Meds

December 31, 2021

1. Which of the following diseases are transmitted by human body louse?
 - A. Epidemic typhus, Relapsing fever, Trench fever
 - B. Yellow fever, Dengue fever, Bubonic plague
 - C. Plague, Endemic typhus, Cat scratch disease
 - D. Q fever, Typhoid fever, Cholera
 - E. Lyme disease, Rockymountain fever, Scrub typhus

ANSWER: A

The Principal Rickettsia Diseases in Humans

Organism	Disease	Arthropod vector	Vertebrate reservoir	Clinical severity	Geographic distribution
I. Typhus					
<i>R. prowazekii</i>	Epidemic typhus Recrudescence typhus (Brill-Zinsser disease)	Louse	Human ^b	++	Africa, South America
<i>R. typhi</i>	Endemic/murine typhus	Flea	Rodents	-	worldwide
<i>Orientia tsutsugamushi</i>	Scrub typhus	Mite ^a	Rodents	++	Far East

^aVertically transmitted in arthropod.

^bNon-human vertebrates are possibly also involved.

III. Others					
Organism	Disease	Arthropod vector	Vertebrate reservoir	Clinical severity	Geographic distribution
<i>Coxiella burnetii</i>	Q fever/Atypical pneumonia	None	Sheep, goats, cattle	+	worldwide
<i>Bartonella henselae</i>	Cat-Scratch fever	none	Cats and dogs		
<i>Bartonella quintana</i>	Trench fever	Louse	Human	+	Asia, Africa, Central and South America ^d
<i>Ehrlichia chaffeensis</i> ^e	Fever (ehrlichiosis)	Tick	?	+	USA, Japan (<i>E. sennetsu</i>)

^dMultiply extracellularly; 1 million soldiers infected in the First World War.

^eIsolated at Fort Chaffee, Arkansas; parasitizes lymphocytes, monocytes, neutrophils.

II. Spotted fever^c					
Organism	Disease	Arthropod vector	Vertebrate reservoir	Clinical severity	Geographic distribution
<i>R. rickettsii</i>	Rocky Mountain spotted fever	Tick ^a	Dogs, rodents	+	Rocky Mountain states, eastern USA
<i>R. akari</i>	Rickettsial pox	Mite ^a	Mice	-	Asia, Far East, Africa, USA
<i>R. conorii</i>	Mediterranean spotted fever	Tick	Dogs	+	Mediterranean
<i>Rickettsia japonica</i>	Oriental spotted fever	Tick/mite			
<i>Rickettsia australis</i>	Queensland tick typhus	Tick			Australia

^aVertically transmitted in arthropod

^cOther rickettsiae cause similar tick-borne fevers in Africa, India, Australia.

Epidemic relapsing fever (Louse borne)

Borrelia recurrentis

2. Which of the following clinical samples gives the highest yield for isolation of *Histoplasma capsulatum*?
- A. Skin
 - B. Tissue
 - C. Sputum
 - D. Bone marrow
 - E. Blood

ANSWER: D, for laboratory diagnosis of *Histoplasma capsulatum*, samples can be taken from Sputum, tissue, skin, **bone marrow**, CSF, blood and lymph nodes. Of these, Bone marrow gives the highest yield for isolation of the organism.

Laboratory diagnosis of *Histoplasma capsulatum* include:

- Direct examination: Giemsa / Wright: **Intracellular** & Extracellular yeast cells. Yeast cells inside macrophages – indicates infection with *H. capsulatum*
- Culture: Mould at 25°C (**tuberculate chlamydospore**) & conversion to yeast on an enriched medium at 37°C
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3. A 37-year-old male is having repeated episodes of fever with chills with blood smear positive for malaria since last one year. He was given chloroquine each time; the fever subsided and the parasitemia reaches to undetectable under microscopy. Which of the following drugs would you add with chloroquine this time?
- A. Primaquine
 - B. Quinine
 - C. Mefloquine
 - D. Malarone
 - E. Artemether

ANSWER: A, Repeated episodes/relapses indicates that the malaria is either caused by *P. vivax* or *P. ovale*. After the parasitemia reaches to undetectable under microscopy, Primaquine should be administered along with chloroquine to eliminate hypnozoite stages (if there're any) of plasmodium vivax or ovale in order to prevent relapses since Primaquine is the only agent that can lead to radical cures of the *P. vivax* and *P. ovale* malarias.

4. The part of the motor neuron enervating muscles that has high density of voltage gated Na⁺ channel is:
- A. Collaterals of axon
 - B. Axon hillock
 - C. Nerve terminal
 - D. Soma
 - E. Dendrites

ANSWER: B

5. Which of the following malaria prevention strategy CANNOT be classified under vector control methods?
- A. indoor residual spray
 - B. repellents applied on the skin
 - C. use of Insecticide Treated Net
 - D. environmental management
 - E. Prompt diagnosis and treatment

ANSWER: E

6. Which condition would most likely lead to the most serious immune deficiency disease?
- A. Lack of cytotoxic T-cells
 - B. Lack of Neutrophils
 - C. Lack of IgG
 - D. Lack of B-cells
 - E. Lack of helper T- cells

ANSWER: E

7. Which one of the following analgesic drugs is metabolized to morphine in the liver via the cytochrome P450 enzyme 2D6 (CYP2D6) and is unsafe if administered to ultra-metabolizer children?
- A. Pethidine
 - B. Heroin
 - C. Aspirin
 - D. Cocaine
 - E. Codeine

ANSWER: E

8. Sarah, a 21-years-old medical student at AAU, College of Health Sciences, had lunch at the college lounge. When she returns to attend class in the afternoon, her vision became blurry, and she was forced to miss the class. After a while. She was having trouble swallowing and speaking clearly. Her friends took her to the hospital. The physician observed that Sarah's breathing was becoming labored and he suspected that she might have Botulism. Based on this case, which of the following statement is correct?
- A. Her symptoms are caused by decreased synthesis of acetylcholine by botulinum toxin
 - B. Her symptoms are caused by an increase in the release of acetylcholine at the mother end plate
 - C. Her symptoms are resulted from autoimmune destruction of nicotinic receptors at the motor endplate
 - D. Her breathing becomes labored due to excessive excitation of her respiratory muscles
 - E. Her symptoms are caused by inhibition of the release of acetylcholine from motor neurons at the motor endplate

ANSWER: E

9. Mechanism of action of tetracyclines
- A. Inhibit peptide bond formation
 - B. Interferes with the formation of initiation complex
 - C. Inhibit binding of tRNA to.....
 - D.
 - E.

ANSWER: C, Tetracyclines act by blocking the binding of aminoacyl-tRNA to the acceptor site on the mRNA-ribosome complex.

10. Associated with pleasure
- A. Dopaminergic Substantia nigra
 - B. Cholinergic nucleus of Mynert
 - C. Serotonergic Raphe nucleus
 - D. GABAergic Striatum nucleus
 - E. Mesocorticolimbic dopaminergic system

ANSWER: E

- **Mesocorticolimbic dopaminergic system** - associated with **pleasure, reward, addiction.....**
- **Serotonergic Raphe nucleus** - **mood controlling** modulatory cortical projections of brain stem
- **Dopaminergic Substantia nigra** - involved in various brain functions such as **voluntary movement**

11. Which one of the following brain waves characterize REM sleep

- A. Beta
- B. Alpha
- C. Theta
- D. Delta
- E. Delta max

ANSWER: A, Beta waves are seen in heightened brain activity and REM sleep

• REM sleep (Paradoxical, Desynchronized)

- Rapid roving movement of the eyes
- Shows **dysynchronized beta** rhythm
- Awakening threshold is high, hence it is called **paradoxical sleep**
- Lasts 5-30 min and recurs every 90 min (its duration increases near morning time making

12. The mood controlling modulatory cortical projections of brain stem are

- A. Dopaminergic Substantia nigra
- B. Cholinergic nucleus of Mynert
- C. GABAergic Striatum nucleus
- D. Serotonergic Raphe nucleus
- E. Adrenergic Locus Coeruleus nucleus

ANSWER: D

- **Serotonergic Raphe nucleus** - **mood controlling** modulatory cortical projections of brain stem
- **Dopaminergic Substantia nigra** - involved in various brain functions such as **voluntary movement**
- **Mesocorticolimbic dopaminergic system** - associated with **pleasure, reward, addiction.....**

13. All of the following are related to the neural tube defects (NTDs) that occur in the embryo
EXCEPT:

- A. Meningomyelocele is from a spina bifida cystica type of NTD
- B. Abnormal closure of the neural folds
- C. Can involve the meninges, vertebrae, muscles, and skin.
- D. Splitting of vertebral arches does not occur in spina bifida occulta
- E. Use of folic acid prior to and during pregnancy is a preventive measure.

ANSWER: D, Spina bifida Occulta occurs due to splitting of vertebral arches, i.e. the outer part of some of the vertebrae is not completely closed. The splits in the vertebrae are so small that the spinal cord does not protrude. It has no or only mild signs, which may include a hairy patch, dimple, dark spot or swelling on the back at the site of the gap in the spine.

14. Cranial nuclei that moves muscles of throat, detect ABP, taste from posterior tongue, and controls salivary gland secretion is
- A. Hypoglossal nerve
 - B. Facial nerve
 - C. Trigeminal nerve
 - D. Glossopharyngeal
 - E. Vagus nerve

ANSWER: D, moves muscles of throat, detect ABP (from carotid sinus), taste from posterior one third of the tongue, and controls salivary (parotid) gland secretion

15. In viral meningitis Cerebrospinal fluid (CSF) analysis can be used to assist the diagnosis of viral meningitis. Which one of the following results of a CSF analysis is a most likely scenario in viral meningitis?
- A. Markedly raised protein
 - B. Markedly turbid CSF
 - C. Reduced glucose concentration
 - D. Elevated number of lymphocytes
 - E. Elevated glucose concentration

ANSWER: D

TABLE 7-4. Important CSF Findings Used to Identify Aseptic, Acute Bacterial, and Chronic Meningitides

Test (adult values)	Aseptic Meningitis	Acute Bacterial Meningitis	Chronic (tuberculous and fungal) Meningitis
Opening pressure (60–180 mm H ₂ O)	Normal or slightly elevated	Elevated (>180 mm H ₂ O)	Normal or mildly increased for tuberculous Increased in fungal meningitis
Clarity (clear, colorless)	Clear or slightly turbid	Turbid; may clot	Clear to slightly turbid; if left to stand, a pellicle or faintly visible "spider's web clot" frequently forms
Protein (15–50 mg/dL)	Increased (50–100 mg/dL)	Increased (often >100 mg/dL)	Increased (average 224 mg/dL) for both
Glucose (45–85 mg/dL)	Normal or slightly decreased	Decreased (<45 mg/dL)	Decreased (<45 mg/dL) in tuberculous Sometimes decreased for fungal
CSF:serum glucose ratio (0.6)	>0.6	<0.3	≤0.4

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54

White blood cells (<5/mL)	*Increased (≤500/mm ³); lymphocytes predominate	Increased (often >1000/mm ³), PMNs predominate	Increased (500/mm ³); initially PMNs then lymphocytes predominate in tuberculous; fungal lymphocytes predominate
Gram stain, acid-fast stain, India ink stain	No organisms	Shows organisms in ~75% of untreated cases	Acid-fast positive in about 25–40% of tuberculous meningitis 80–90% of fungal meningitis cases are positive for India ink staining (see Figure 7-3)
Culture	No organisms	Organisms grow in ~85% of untreated cases	Tuberculous meningitis cultures can take up to 4 weeks PCR amplification of bacterial DNA highly specific and much faster Fungal positive within 7 days for >90% of patients

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55

16. Which of the following is FALSE about herniation?

- A. Sub falcine herniation result in compression of anterior cerebral artery.
- B. Tonsillar herniation may result in respiratory failure.
- C. In transtentorial herniation, patients develop contralateral pupillary dilatation and ipsilateral hemiparesis.
- D. Patients with transtentorial herniation end up with compression of the aqueduct.
- E. Tonsillar herniation results in brainstem compression.

ANSWER: C, In transtentorial herniation, patients develop ipsilateral pupillary dilatation/mydriasis/ as well as ipsilateral hemiparesis.

17. A previous healthy 31 years old woman experiences a sudden severe headache and loses consciousness within an hour. An emergent head CT scan reveals extensive subarachnoid hemorrhage at the base of the brain. She is afebrile. A lumbar puncture yields CSF with many red blood cells, but no white blood cells. The CSF protein is slightly increased, but the glucose is normal. Which of the following is the most likely diagnosis?
- A. Hypertensive hemorrhage
 - B. Acute Bacterial Meningitis
 - C. Progressive Multifocal Leukoencephalopathy
 - D. Amyloid arteriopathy
 - E. Ruptured berry aneurysm

ANSWER: E,

- **Subarachnoid hemorrhage** - The most frequent cause is a **rupture of a saccular (berry) aneurysm**. Berry aneurysm (saccular or congenital aneurysm) is the most common type intracranial aneurysm which mostly occur in anterior circulation of circle of willis.
- **Subdural hematoma** -- The **bridging veins** can be torn along their course in dura space and result in subdural hematoma.
- **Epidural hematoma** – commonly occurs due to tearing the **middle meningeal artery**.

18. Which one is the correct sequence of onset of anesthesia?
- A. Motor -- pain -- inhibitory -- excitatory neurons
 - B. Inhibitory -- excitatory -- pain -- motor neurons
 - C. Excitatory -- pain -- motor -- inhibitory neurons
 - D. Inhibitory -- pain -- motor -- excitatory neurons
 - E. Pain -- inhibitory -- excitatory -- motor neurons

ANSWER: E

19. If you observe cottony matrix topped by small black balls in the external auditory canal during otoscope examination, the most likely organism is: -
- A. Pseudomonas aeruginosa
 - B. Aspergillus niger
 - C. Candida albicans
 - D. Streptococcus pneumoniae
 - E. Aspergillus fumigatus

ANSWER: B

- Otitis externa, also called swimmer's ear, is inflammation of the ear canal.
- Cottony matrix topped by small **black** balls – Characteristic of *Aspergillus niger*
- Otitis externa with **blue green** pigment -- *Pseudomonas aeruginosa*
- Commonest cause of Otitis externa -- *Pseudomonas aeruginosa*
- Commonest bacterial cause of **Otitis media** -- *Streptococcus pneumoniae*
-

20. Lack of fusion of paramesonephric ducts results in

- A. Uterus arcuatus
- B. Vaginal atresia
- C. Uterus didelphys
- D. Uterus atresia
- E.

ANSWER: C, Uterus didelphys – double uterus.

21. Which hormone prevents precocious puberty/ delays puberty?

- A. Leptin
- B. Melatonin
- C. Parathyroid hormone
- D. Cortisol
- E. Growth hormone

ANSWER: B

22. Which hormone controls energy metabolism by adipose/fat deposition

- A. Leptin
- B. Melatonin
- C. Growth hormone
- D. Prolactin
- E. Thyroxine

ANSWER: A

23. Menstruation is caused by
- A. Withdrawal of sex hormones
 - B. Estrogen
 - C. Progesterone
 - D. Follicle stimulating hormone
 - E. Luteinizing hormone surge

ANSWER: A, Menstruation is caused by withdrawal of sex hormones (withdrawal of estrogen and progesterone)

24. Which layer of the uterus undergoes profound changes(sheds) during menstrual cycle?
- A. Basal layer of endometrium
 - B. Myometrium
 - C. Functional layer of endometrium
 - D. Perimetrium
 - E. Epimetrium

ANSWER: C

25. Krukenberg tumor indicates
- A. Metastatic gastric carcinoma to ovaries
 - B. Metastatic gastric carcinoma to the umbilicus
 - C. Metastatic esophageal carcinoma to the lung
 - D. Metastatic esophageal carcinoma to the pleura
 - E. Metastatic gastric carcinoma to supraclavicular.....

ANSWER: A

- Gastric carcinomas frequently metastasize to the supra clavicular sentinel (**Virchow**) node as the first clinical manifestation
- The tumor can also metastasize to the **periumbilical region** – **Sister Mary Joseph nodule**
- Metastatic adenocarcinoma to the ovaries is called **Krukenberg tumor**

26. Which one of the following factors increases GFR?

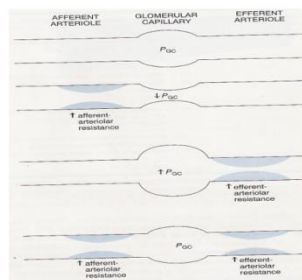
- A. Renal efferent vasoconstriction
- B. Increased voiding of urine
- C. Renal afferent vasoconstriction
- D. Increase Bowman's interstitial capillary pressure
- E. Sympathetic stimulation of renal afferent arteriole

ANSWER: A

Diameter of the renal arterioles also regulate GFR

1. **Afferent** arteriolar constriction : **Decreases GFR** because constriction increases arteriolar resistance that causes decreased blood flow to the Glomerulus and this in turn decreases glomerular pressure that results in **decreased GFR**

2. **Efferent** arteriolar constriction: **increase GFR**
Constriction of the efferent arterioles increase resistance to blood flow and this effect results in **increased GFR**



27. Which of the following embryonic GI structure **do not** undergo rotation during its development?

- A. Stomach
- B. Duodenum
- C. Hind gut
- D. Intestinal loop
- E. Ventral pancreas

ANSWER: C

28. Which statement correctly defines the proximal tubule (PT)?
- A. Most of the glucose and amino acids are reabsorbed in the PT
 - B. ADH acts on proximal cells to reabsorb water
 - C. PT contain JGC (juxtaglomerular cells) which secrete renin
 - D. The filtrate in PT is hypo-osmotic
 - E. The filtrate in PT is iso-tonic

ANSWER: E, but choice A can also be an answer.

29. Which one of the following is NOT a function of a nephron

- A. Site of urine storage
- B. Site of filtration
- C. Excretion of metabolites
- D.
- E.

ANSWER: A, the nephron filters blood and forms urine. Urinary bladder is the site of urine storage.

30. Which of the following is NOT TRUE about cystic diseases of the kidney?
- A. Polycystic kidney diseases are important causes of chronic renal failure
 - B. The clinical presentation may mimic renal tumors
 - C. Autosomal dominant polycystic kidney disease manifests clinically in infancy
 - D. Solitary cysts are usually asymptomatic
 - E.

ANSWER: C

- Autosomal **dominant** polycystic kidney disease manifests clinically in **adulthood**
- Autosomal **recessive** polycystic kidney disease manifests clinically in **infancy**

31. GI hemorrhage & GI obstruction is associated with which of the following heavy metal
- A. Lead
 - B. Mercury
 - C. Cadmium
 - D. Iron
 - E. Arsenic

ANSWER: D

32. Identification of khat toxicity in khat chewers needs

- A. Experimental study in animal model
- B. Epidemiological survey
- C. Experimental study in human
- D. Clinical investigation
- E. Occupational investigation

ANSWER: B

33. A diagnosis of Meckel's diverticulum was made. This condition is associated with which of the following?

- A. Nonrotation of the midgut.
- B. Herniation of the intestines.
- C. Duplication of the intestine.
- D. Subhepatic cecum and appendix
- E. Remnant of the yolk stalk.

ANSWER: E

34. Endoscopic examination revealed tortuous dilated submucosal veins of distal esophagus. This is typical for

- A. Barret esophagus
- B. Esophageal varices
- C. Caput medusa
- D. Adenocarcinoma
- E. Mallory weiss tear

ANSWER: B

35. The preferred non-invasive test to confirm H. pylori eradication is

- A. a whole-blood antibody-detection test.
- B. a urea breath test.
- C. a stool antigen test.
- D. a serologic antibody detection test.
- E. urea....

ANSWER: B

36. Which of the ff best describes the relationship between parasite and host?

- A. The most successful parasites are those that can kill their hosts
- B. Facultative parasites are opportunistic in their hosts
- C. The association between parasites and hosts is a symbiotic mode of life
- D. Parasites induce humoral immune responses, but not cell mediated immunity
- E. Parasites have a higher reproductive potential than hosts

ANSWER: E

37. All of the following increase gastric acid secretion EXCEPT

- A. Increased somatostatin production
- B. Increased gastrin production
- C. Zollinger-Ellison syndrome
- D. Increased histamine production
- E. Increased vagal stimulation

ANSWER: A

38. A drug that will decrease the amount of acid secreted by binding to and inhibiting an H⁺/K⁺ ATPase on the luminal surface of parietal cells.

- A. Octreotide
- B. Misoprostol
- C. Atropine
- D. Cimetidine
- E. Omeprazole

ANSWER: E

39. You have two patients. One has nausea and vomiting, the other with gastroparesis. What's the most suitable drug for both patients

- A. Loperamide
- B. Metoclopramide
- C. Promethazine
- D. Ondansetron
- E. Meclizine

ANSWER: B

40. Select the statement which is NOT CORRECT with regard to the histology of the heart.
- A. Purkinje cells travel in subendocardial layer of the endocardium to the apex of the heart.
 - B. Pectinate muscles are the sites of attachment of chordae tendinae.
 - C. Cardiac muscle is more vascularized and has more mitochondria than does skeletal muscle.
 - D. Epicardium is a serous covering of the heart, forming the visceral layer of the pericardium.
 - E. Endocardium is the thickest in atria and the thinnest in ventricles, especially the left ventricle.

ANSWER: B

41. Useful landmark to count ribs.....
- A. Sternal angle
 - B. Xiphoid process
 - C. Jugular notch
 - D.
 - E.

ANSWER: A

42. Correct match between drugs and their mechanism of action, site of action
- A. Captopril -- Angiotensin converting enzymes inhibitor in lung tissue.....
 - B. Verapamil – calcium channel blocker in blood vessels
 - C.
 - D.
 - E.

ANSWER: A

- Verapamil – calcium channel blocker in the heart, not in blood vessels

43. What structures work to prevent prolapse (eversion) of the atrioventricular valves?
- A. Pectinate muscles
 - B. Autorhythmic cells
 - C. Trabeculae carneae
 - D. Papillary muscles
 - E.

ANSWER: D, **chorda tendinea** and papillary muscles work together to prevent prolapse or eversion of the atrioventricular (mitral and tricuspid) valves

44. Neutral amino acid

- A. Glutamate
- B. Valine
- C. Aspartate
- D.
- E.

ANSWER: B

45. Parietal layer covering pleura is derived from.

- A.
- B. Ectoderm
- C. Endoderm
- D. Intraembryonic mesoderm
- E. Somatic extraembryonic mesoderm

ANSWER: E

46. Endochondral bone ossification includes the following, except:

- A. Chondrocytes changing to osteocytes.
- B. Proliferation of cartilage cells.
- C. Calcification of the cartilage matrix.
- D. Interruption of diffusion to the cartilage at the area of primary ossification.
- E. A cartilage model

ANSWER: A

47. TRUE

- A. Collagen is characterized by high tensile strength and flexibility.
- B.
- C.
- D.

E.

ANSWER: A

48. Iron adverse effect in GI

A. Delirium,.....

B. Necrotizing fascitis,.....

C.

D.

E.

ANSWER: B

49. One of the following muscles is not innervated by posterior interosseous nerve?

A. Extensor carpi radialis brevis

B. Extensor pollicis longus

C. Extensor digiti minimi

D. Abductor pollicis longus

E.

ANSWER: A

50. Which of the following enzymes removes RNA primers and adds DNA nucleotides into a growing strand of DNA

A.

B.

C. DNA Polymerase III

D.

E.

ANSWER: C

51. Not involved in proton pump..... energy generation

A. Complex I

B. Complex II

C. Complex III

D. Complex IV

E. Complex V

ANSWER: B

52. A type of necrosis in which the underlying tissue structure is preserved for at least several days

- A. Liquefactive necrosis
- B. Coagulative necrosis
- C. Caseous necrosis
- D. Gangrenous necrosis
- E. Fat necrosis

ANSWER: B

53. An infant born with mental retardation, slanted palpebral features,, duodenal atresia....

- A. Trisomy 21
- B. Trisomy 18
- C. Trisomy 13
- D. Turner syndrome
- E. Klinefelter syndrome

ANSWER: A, Trisomy 21 or Down syndrome

Trisomy 21 → mental retardation, slanted palpebral fissures, hypertelorism, macroglossia, atrial septal defect (ASD), duodenal atresia, early-onset Alzheimer disease, and multiple

visceral anomalies

Down syndrome

Bilious vomiting, "double bubble," associated with Trisomy 21

Duodenal atresia

54. Which is used for opsonization and killing

- A. C3b, Fc portion of IgG
- B.
- C.
- D.

E.

ANSWER: A

55. Which statement is true

- A. Elastic cartilage.....
- B. Fibrocartilage is present in regions subjected to high tension.
- C. Hyaline cartilage.....
- D.
- E.

ANSWER: B

56. Which of the following is the most common benign soft tissue tumor of adulthood?

- A. Fibromatosis
- B. Neurofibroma
- C. Lipoma
- D. Nodular Fascitis
- E. Rhabdomyoma

ANSWER: C

57. Which statement about slow twitch fibers is correct?

- A. Scarce mitochondria, Extensive capillary supply, low concentrations of myoglobin and Fatigue resistant
- B. Abundant mitochondria, Extensive capillary supply, low concentrations of myoglobin and fatigue resistant
- C. Abundant mitochondria, Extensive capillary supply, High concentrations of myoglobin and Fatigue resistant
- D. Scarce mitochondria, Extensive capillary supply, High concentrations of myoglobin and Fatigue resistant
- E. Abundant mitochondria, Extensive capillary supply, High concentrations of myoglobin and easy fatigue

ANSWER: C

58. Which of the following is most likely associated with rigor mortis

- A. Digestion of muscle proteins by lysosomes
- B. Depletion of ATP
- C. Increased myosin ATPase activity
- D. Damage to actin and myosin filaments
- E. Rapid sequestration of Ca^{2+} back to the SR

ANSWER: B

59. The activity of muscle glycogen synthesis in resting muscle is increased by the action of which of the following?

- A. Norepinephrine
- B. Cortisol
- C. Epinephrine
- D. Glucagon
- E. Insulin

ANSWER: E

60. In which layer of the cerebrum are major synapses found?

- A. Inner granular layer
- B. Inner pyramidal layer
- C. Plexiform/molecular layer
- D. Outer granular layer
- E. Outer pyramidal

ANSWER: C

61. Major cause of blindnessaround fast running rivers

- A. Onchocerciasis
- B. Elephantiasis
- C. Loa Loa
- D. Ascariasis
- E. Leishmaniasis

ANSWER: A

62. Doesn't have.....zone.....B lymphocytes

- A. Thymus
- B. Spleen
- C. Lymph nodes
- D.
- E.

ANSWER: A

63. False about urinary tract infections

- A. Males have shorter urethra
- B.
- C.
- D.
- E.

ANSWER: A

64. Malaria parasite can cause which type of Jaundice?

- A. Obstructive
- B. Hemolytic
- C. Congenital Hyperbilirubinemia
- D. Post hepatic
- E. Hepatocellular

ANSWER: B

65. Amiodarone side effects

- A. Pulmonary fibrosis and thyrotoxicosis
- B.
- C.
- D.
- E.

ANSWER: A

66. Treatment for Nocturnal Asthma

- A. Salbutamol
- B. Salmeterol
- C. Betamethasone
- D. Montelukast
- E. Omalizumab

ANSWER: B

67. When a muscle is at rest, the main source of energy is

- A. Blood glucose
- B. Serum fatty acid
- C. Muscle glycogen
- D. Liver glycogen
- E. Ketone bodies

ANSWER: B

68. Which of the following statements best describes motor units?

- A. Are the same at neuromuscular junctions.
- B. Consists of a muscle fiber and all the nerves that supply it
- C. Are found only in cardiac muscle
- D. Are largest in muscles involved for delicate movement
- E. Consists of a motor neuron and all the muscle fibers it supplies

ANSWER: E

69. The most important feature that helps distinguish malignant from benign tumors is

- A. Rate of growth
- B. Local invasion
- C. Metastasis
- D. Microscopic features
- E. Clinical and gross features

ANSWER: C

70. The main predisposing factor for the development of cancer in developing nations is:

- A. Chronic infections
- B. Radiation
- C. Tobacco
- D. Alcohol
- E. Occupational hazards

ANSWER: A

71. Which of the following represents the most common form of hepatic porphyria?

- A. ALA dehydratase deficient porphyria
- B. Acute intermittent porphyria
- C. Porphyria cutanea tarda type I
- D. Hereditary coproporphyria
- E. Variegate porphyria

ANSWER: B

72. True about systemic mycosis

- A. Are caused by opportunistic pathogens
- B. Causative agents are introduced by trauma
- C. Causative agents are all dimorphic
- D.
- E.

ANSWER: C

73. During the development of the heart

- A. Left horn of sinus venosus is incorporated into the wall of the right atrium
- B. Cavitation of the ventricular walls forms the trabeculae-carneae
- C. Bulbuscordis forms the trabeculated part of the right ventricle
- D. Intra-abdominal part of the umbilical vein becomes the median umbilical ligament
- E. Sinoatrial nodes are derived from the atrioventricular canal

ANSWER: B

74. Neurogenic shock is best managed with

- A. IV fluid and vasodilators
- B. Vasoconstrictors
- C. Fluid drainage
- D. Adrenaline
- E. Prazosine

ANSWER: B

75. Commonest cause of esophagitis in immune competent individuals is:

- A. Achalasia
- B. Reflux disease
- C. Congenital
- D. Chemicals
- E. Infections

ANSWER: B

76. Identify the correct alternative about bacterial pathogenicity and virulence factors:

- A. Endotoxins are integral part of cell wall which are released before bacterial lysis
- B. Pathogenicity is defined as extra-chromosomal pieces of DNA
- C. Diphtheria toxin activates elongation factor 2
- D. Superantigens activate T cells by binding to TCR and MHCII molecule on an APC
- E. Enterotoxins are endotoxins acting on gastrointestinal tract

ANSWER: D

77. One of the following is NOT a cholinergic toxidrome

- A. Lacrimation
- B. Urination
- C. Bradycardia
- D. Diaphoresis
- E. Hypotension

ANSWER: E, HYPOTENSION is NOT a cholinergic toxidrome.

- Diaphoresis means excessive sweating that can result from overstimulation of sweat glands by Acetylcholine. Sweat glands are innervated by sympathetic cholinergic nerve endings.

78. Select the antimalarial agent which is most commonly associated with acute hemolytic reaction in patients with glucose-6-phosphate dehydrogenase deficiency

- A. Artesunate
- B. Chloroquine
- C. Primaquine
- D. Quinine
- E. Mefloquine

ANSWER: C

79. All of the following are features of the right atrium, EXCEPT:

- A. Sino atrial node
- B. Fossa ovalis
- C. Papillary muscles
- D. Pectinate muscles
- E. Opening of coronary sinus

ANSWER: C

80. A muscle that is chiefly responsible for a particular movement:

- A. Synergist
- B. Prime mover
- C. Extensor
- D. Flexor
- E. Antagonist

ANSWER: B

81. Which one of the following is used as a biomarker for myocardial infarction

- A. ALT
- B. Troponin
- C. C-reactive protein
- D. HbA1c
- E. Bilirubin

ANSWER: B

82. Which one is most indicative for liver abnormalities

- A. Alanine transaminase
- B. Acid phosphatase
- C. Aspartate aminotransferase
- D. Glutamate aminotransferase
- E. Alkaline phosphatase

ANSWER: A

83. For vaccination against Mycobacterial diseases, the most important aspect of the immune response to be stimulated is

- A. Proliferation of cytotoxic T cells
- B. Antibody mediated immunity
- C. Production of antibody in the gut lumen
- D. Macrophage-activating cell-mediated immunity
- E. Activation of neutrophils

ANSWER: D

84. A 12 year-old female patient presented with right ovarian mass. On further work up it was discovered that she has a primary ovarian tumor with adult brain tissue, cartilage mixed with other adult tissues. What is the correct diagnosis?

- A. Granulosa cell tumor
- B. Krukenberg tumor
- C. Mature teratoma
- D. Serous borderline tumor
- E. Choriocarcinoma

85. Which of the following provides information about the variation in sensitivity of a drug within a population?

- A. Maximal efficacy
- B. Therapeutic index
- C. Drug potency
- D. Grade dose-response curve
- E. Quantal dose-response curve

ANSWER: E

86. Atropine is not administered for.....

ANSWER: Atropine is not administered for Asthma

Atropine is used for cholinergic poisoning. See the table below for other uses of muscarinic blockers.

	Drug	Therapeutic uses
Muscarinic blockers	Cyclopentolate Tropicamide Atropine*	In ophthalmology, to produce mydriasis and cycloplegia prior to refraction
	Atropine*	To treat spastic disorders of the GI and lower urinary tract To treat organophosphate poisoning To suppress respiratory secretions prior to surgery
	Scopolamine	In obstetrics, with morphine to produce amnesia and sedation To prevent motion sickness
	Ipratropium	Treatment of asthma
	Nicotine	None
Ganglionic blockers	Mecamylamine	Treatment of moderate to severe hypertension

Figure 5.7 Summary of cholinergic antagonists. *Contraindicated in narrowangle glaucoma. GI = gastrointestinal.

87. A patient ate..... what is the most likely cause of his symptoms?

A. He ate Tyramine containing diet

B.

ANSWER: A

88. A function of parasympathetic nervous system

A. Lacrimation

B. Vasoconstriction

C.

ANSWER: A

89. Basal ganglia is responsible for

A. voluntary movement

B.

ANSWER: A

90. Which of the following is a feature of Mycobacterium avium intracellulare complex infection?

A. Organisms are found within macrophages (foam macrophages)

B. Infection usually occurs in immunocompetent individuals

C.

D.

E.

ANSWER: A

91. Paralysis of facial muscles is associated with lesion in the

A. Inferior part of premotor cortex

B.

C.

ANSWER: A

92. Which of the following statement is true regarding TB

A. Risk of developing TB disease is highest after two years of infection

B. People who are exposed to TB may or may not develop TB infection

C.

ANSWER: B

93. False statement

A. Glans is derived from corpora cavernosum

B.

C.

ANSWER: A

94. Bone tumor that doesn't grow on epiphysis.....

A. Osteosarcoma

B.

C.

D.

E.

ANSWER: A

95. A problem of primary hemostasis

- A. Deficiency of von willebrand factor
- B. Hemophilia
- C.
- D.
- E.

ANSWER: A

96. Which of the following anti cancer drugs act by cross linking DNA

- A. Vinbasting
- B. Cisplatin
- C.
- D.
- E.

ANSWER: B

97. Nerve injured during thyroidectomy

- A. Recurrent laryngeal nerve
- B. Superior laryngeal nerve
- C. External laryngeal nerve
- D.
- E.

ANSWER: A

98. Which of the following statement is True?

- A. Ventricular repolarization generates the P wave
- B. The QRS complex coincides with ventricular repolarization
- C. Atrial repolarization generates the T wave
- D. The P-R interval is normally.....
- E.

ANSWER: __

99. A question about a tumor suppressor gene in which both of its alleles mutated and in relation to osteosarcoma of femur...

ANSWER: Ocular retinoblastoma

- 100. Beta blockerpropranolol
- 101. Crescentic glomerulonephritis
- 102. Bcr-abl gene
- 103. Glycemic index and glycemic load
- 104. VLDL
- 105. dilated cardiomyopathy
asymptomatic ventricular hypertrophy
- 106. glycine
- 107. Nuclear bag and nuclear chain
- 108. Schistosoma cancer